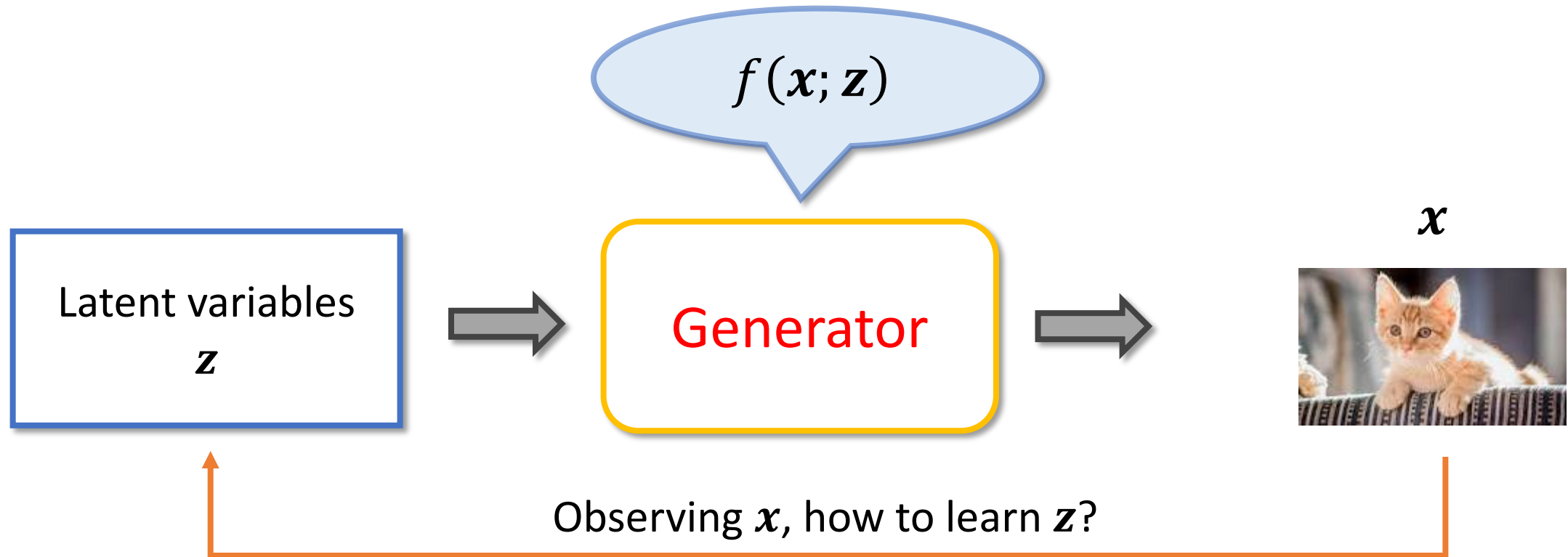


CS 57800 – Statistical Machine Learning

Lecture 2.5 Latent Variables Learning (II)

https://ruizhezhang.com/course_fall_2026.html

Latent Variable Models

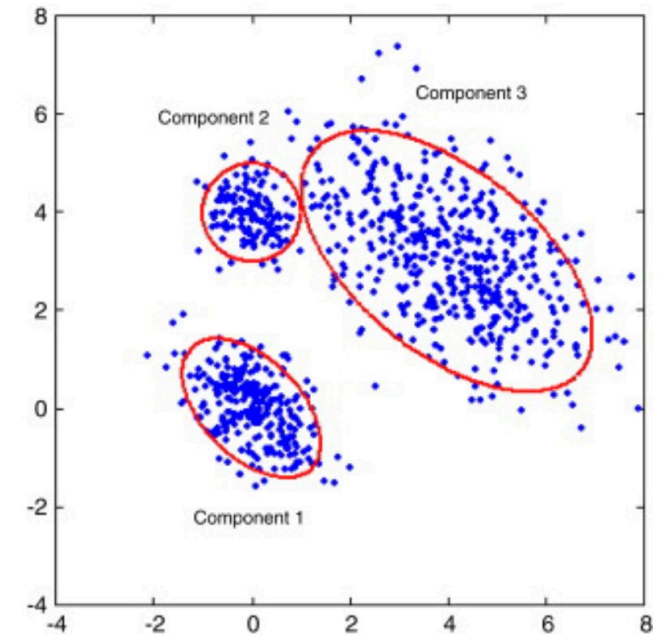


Outline

- Matrix Completion
- Method of Moments

Two Canonical Examples

- **Mixture of Gaussians**
 - Each point is generated by an (unknown) cluster
- **Topic models**
 - “bag of words” model for documents
 - documents have one (or more) topics



What is the statistical efficiency of the estimator we find?

Heuristics: k -means, expectation–maximization (EM), ...

Two Canonical Examples

(spherical) Mixture of Gaussians

- k means: $\mu_1, \dots, \mu_k \in \mathbb{R}^d$
- Sample cluster $i \in [k]$ with probability w_i
- Observe x , with spherical noise,
$$x = \mu_i + \eta, \quad \eta \sim \mathcal{N}(0, \sigma_i^2 I)$$
- Dataset: multiple points

(single) Topic Model

- k topics: $\mu_1, \dots, \mu_k$
- Sample topic $i \in [k]$ with probability w_i
- Observe m (**exchangeable**) words
$$x_1, x_2, \dots, x_m \sim \text{i.i.d. } \mu_i$$
- Dataset: multiple m -word documents

How to learn the parameters $\mu_1, \dots, \mu_k, w_1, \dots, w_k$ (and σ_i^2)?

Detour: Topic Model

A concrete example



Documents with explicit topics
e.g., newspaper, magazine, ...

bag-of-words



	$w_1 = 0.6$	$w_2 = 0.4$
Word	Topic 1: “sports”	Topic 2: “science”
team	0.5	0.05
game	0.35	0.1
experiment	0.1	0.5
data	0.05	0.35

$$\mu_1 = \begin{bmatrix} 0.5 \\ 0.35 \\ 0.1 \\ 0.05 \end{bmatrix}$$

$$\mu_2 = \begin{bmatrix} 0.05 \\ 0.1 \\ 0.5 \\ 0.35 \end{bmatrix}$$

Detour: Topic Model

A concrete example



Documents with explicit topics
e.g., newspaper, magazine, ...

bag-of-words



Word	$w_1 = 0.6$	$w_2 = 0.4$
	Topic 1: "sports"	Topic 2: "science"
team	0.5	0.05
game	0.35	0.1
experiment	0.1	0.5
data	0.05	0.35

Generation:

$$\Pr[x_1 = \text{"game"}, x_2 = \text{"data"} | \text{Topic 1}] = 0.35 \cdot 0.05$$



T1



T1



T2



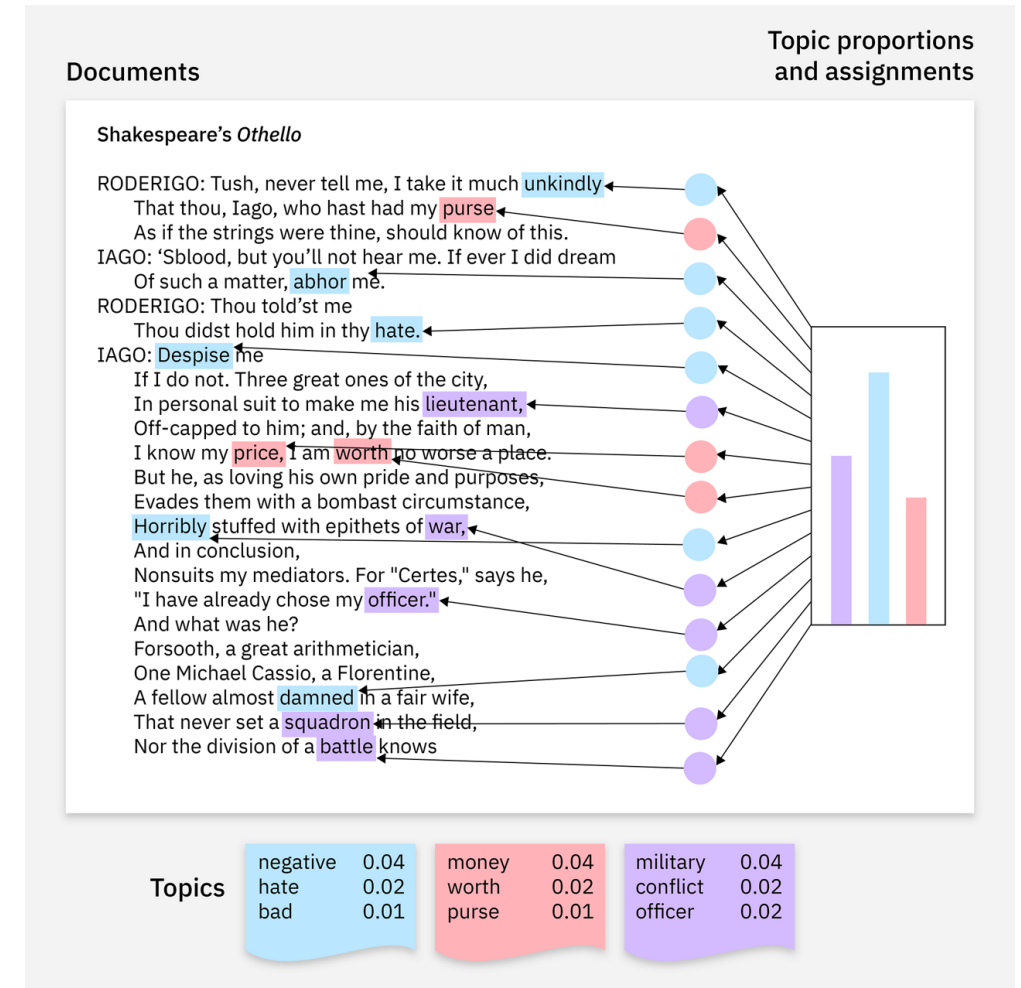
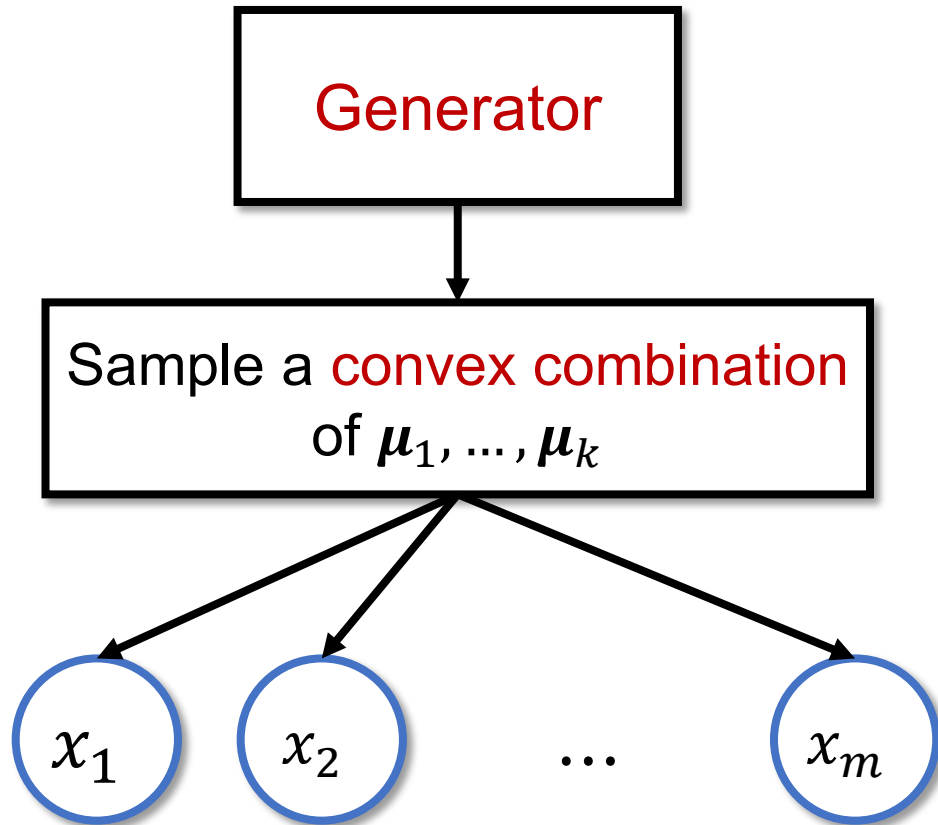
T1

$$\mu_1 = \begin{bmatrix} 0.5 \\ 0.35 \\ 0.1 \\ 0.05 \end{bmatrix}$$

$$\mu_2 = \begin{bmatrix} 0.05 \\ 0.1 \\ 0.5 \\ 0.35 \end{bmatrix}$$

Detour: Topic Model

Real-world documents are mixtures of topics



Two Canonical Examples

(spherical) Mixture of Gaussians

- k means: $\mu_1, \dots, \mu_k \in \mathbb{R}^d$
- Sample cluster $i \in [k]$ with probability w_i
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(single) Topic Model

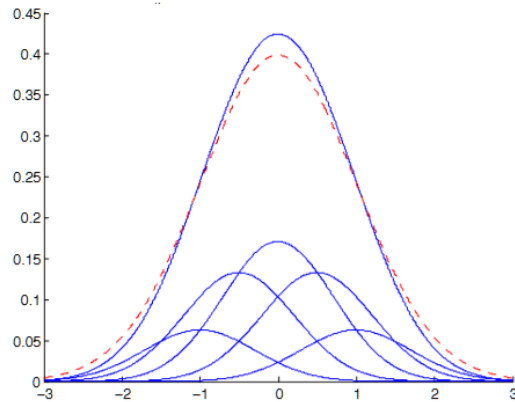
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- Dataset: multiple m -word documents

How to learn the parameters $\mu_1, \dots, \mu_k, w_1, \dots, w_k$ (and σ_i^2)?

Limits of Learning

Information-theoretical lower bound:

- **Moitra-Valiant '11:** need exponential in k many samples to learn a GMM with k components, even in 1-d



Fun fact: A discrete Gaussian mixture is almost Gaussian

Let $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. Then

$$\frac{1}{Z} \sum_{i \in \mathbb{Z}} \phi(i/k) \cdot \mathcal{N}(i/k, 1) \approx_{e^{-\Omega(k^2)}} \mathcal{N}(0, 2)$$

Computational lower bound:

- **Arora-Ge-Moitra '12:** computing the maximum likelihood estimation for learning the parameter of a topic model, more specifically, a [Latent Dirichlet Allocation \(LDA\)](#) model, is **NP-hard**

Limits of Learning

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Computational lower bound:

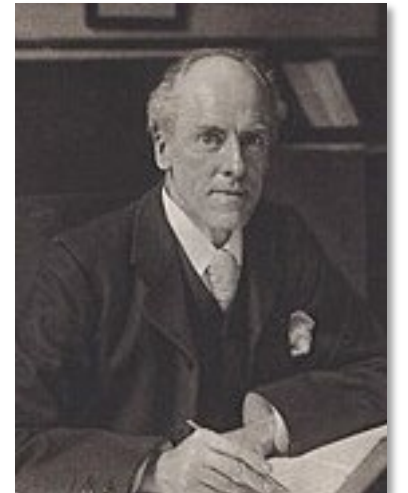
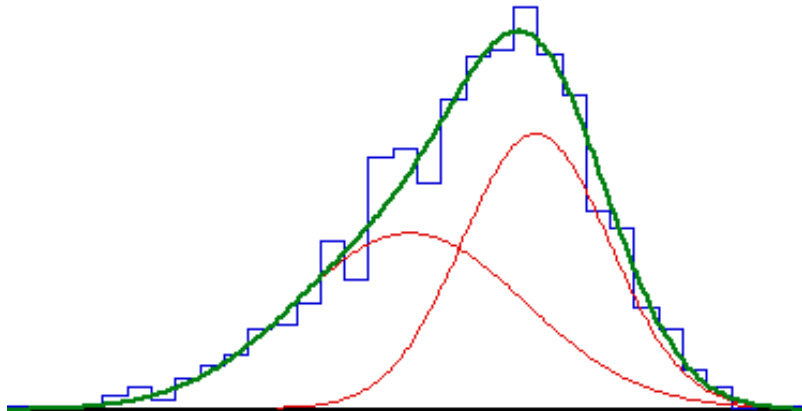
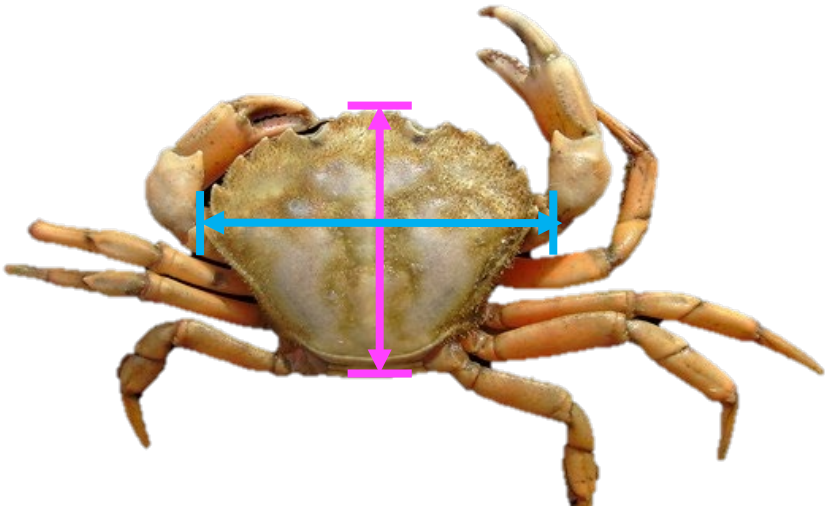
- **Arora-Ge-Moitra '12**: computing the maximum likelihood estimation for learning the parameter of a topic model, more specifically, a [Latent Dirichlet Allocation \(LDA\)](#) model, is **NP**-hard

Are there statistically and computationally efficient methods?

- Under what assumptions and models?
- How general?

Method of moments

- A simple approach based on linear algebra
 - For non-convex problems
- Extensions to richer settings
 - LDA, Hidden Markov model (HMM), etc.
- In high dimensions, GMMs and topic models can be efficiently learnable
- Pearson's crab data (1894)



Karl Pearson
(1857-1936)

The Method of Moments

- GMM moments:

$$\mathbb{E}[\mathbf{x}], \quad \mathbb{E}[\mathbf{x}\mathbf{x}^\top], \quad \mathbb{E}[\mathbf{x} \otimes \mathbf{x} \otimes \mathbf{x}], \quad \dots$$

- Topic model moments:

$$\Pr[x_1], \quad \Pr[x_1, x_2], \quad \Pr[x_1, x_2, x_3], \quad \dots$$

- These moment data can be estimated from observed data
- **Identifiability problem:** with **exact moments**, which order suffices?
 - Can we design efficient algorithms?

Kronecker product

For $A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{p \times q}$,

$$(A \otimes B)_{ik,j\ell} = A_{ij}B_{k\ell}$$

The Method of Moments

Setups

- There are k clusters/topics and d dimensions/words. $d \gg k$
- For GMMs,
 - $\mathbf{x} = \boldsymbol{\mu}_i + \boldsymbol{\eta}$, $\boldsymbol{\eta} \sim \mathcal{N}(0, \sigma_i^2 I)$
 - $\mathbb{E}[\mathbf{x} | \text{cluster } i] = \boldsymbol{\mu}_i$
- For topic models,
 - Use one-hot encoding for words: $\mathbf{x}_1 = [0 \quad 1 \quad 0 \quad \dots]^\top$
 - The $\boldsymbol{\mu}_i$'s are probability vectors
 - $\Pr[x_1 | \text{topic } i] = \mathbb{E}[\mathbf{x}_1 | \text{topic } i] = \boldsymbol{\mu}_i$

The Method of Moments

First moment

GMM

$$\mathbb{E}[\mathbf{x}] = \sum_{i=1}^k w_i \boldsymbol{\mu}_i$$

Topic model

$$\Pr[x_1] = \sum_{i=1}^k w_i \boldsymbol{\mu}_i$$

Not identifiable: d nums, $dk + k$ vars

The Method of Moments

Second moment

GMM

$$\begin{aligned}\mathbb{E}[\mathbf{x} \otimes \mathbf{x}] &= \mathbb{E}[(\boldsymbol{\mu}_i + \boldsymbol{\eta}) \otimes (\boldsymbol{\mu}_i + \boldsymbol{\eta})] \\ &= \mathbb{E}[\boldsymbol{\mu}_i \otimes \boldsymbol{\mu}_i] + \mathbb{E}[\boldsymbol{\eta} \otimes \boldsymbol{\eta}] \\ &= \sum_{i=1}^k w_i \boldsymbol{\mu}_i \otimes \boldsymbol{\mu}_i + \sigma^2 I\end{aligned}$$

- A full-rank matrix

Topic model

By **exchangeability**,

$$\begin{aligned}\Pr[x_1, x_2] &= \mathbb{E}[\mathbb{E}[\mathbf{x}_1 | \text{topic}] \otimes \mathbb{E}[\mathbf{x}_2 | \text{topic}]] \\ &= \sum_{i=1}^k \boldsymbol{\mu}_i \otimes \boldsymbol{\mu}_i\end{aligned}$$

- A low-rank matrix!

Still not identifiable

The Method of Moments

Second moment

Topic model

$$\Pr[x_1, x_2] = \sum_{i=1}^k w_i \boldsymbol{\mu}_i \otimes \boldsymbol{\mu}_i$$

Model A: $w_1 = 1/3, \boldsymbol{\mu}_1 = \begin{bmatrix} 1/6 \\ 5/6 \end{bmatrix}$ and $w_2 = 2/3, \boldsymbol{\mu}_2 = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}$

Model B: $w_1 = 2/3, \boldsymbol{\mu}_1 = \begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix}$ and $w_2 = 1/3, \boldsymbol{\mu}_2 = \begin{bmatrix} 5/6 \\ 1/6 \end{bmatrix}$

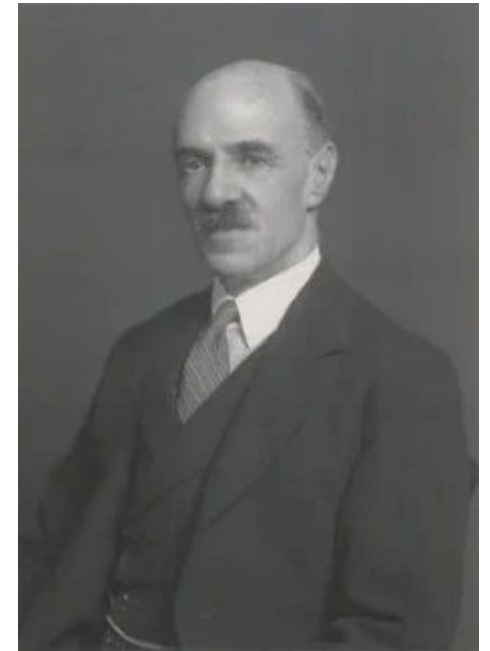
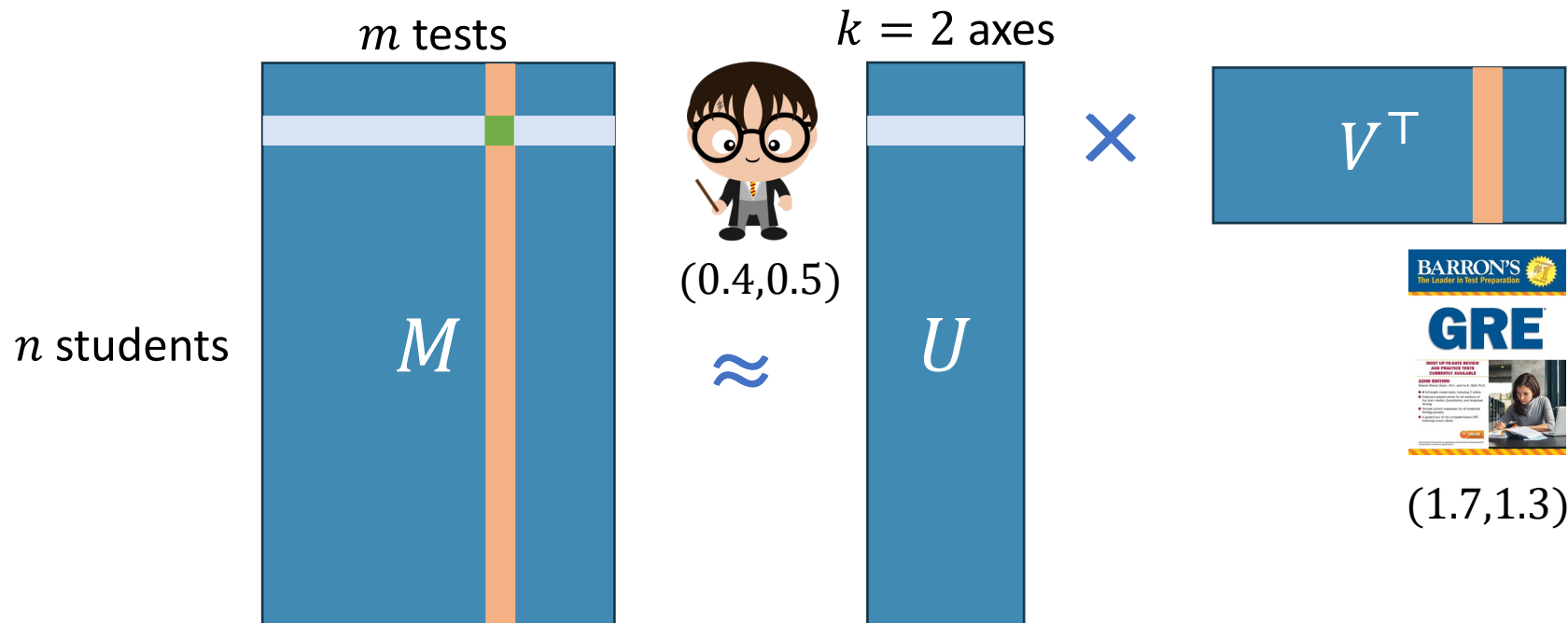
$$\Pr[x_1] = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

$$\Pr[x_1, x_2] = \begin{bmatrix} 11/36 & 7/36 \\ 7/36 & 11/36 \end{bmatrix}$$

Detour: the Rotation Problem

Factor analysis is a statistical method, pioneered by Charles Spearman, that explains observed correlations among many variables by modeling them as combinations of a few underlying latent factors

Suppose a psychologist has the hypothesis that there are two kinds of intelligence, “verbal” and “mathematical”.

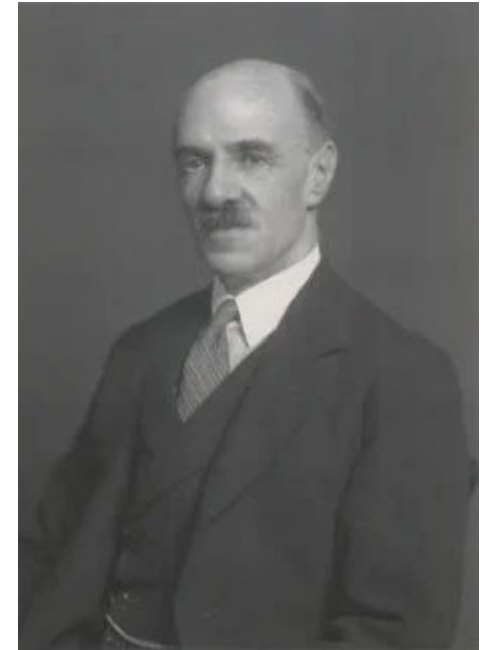
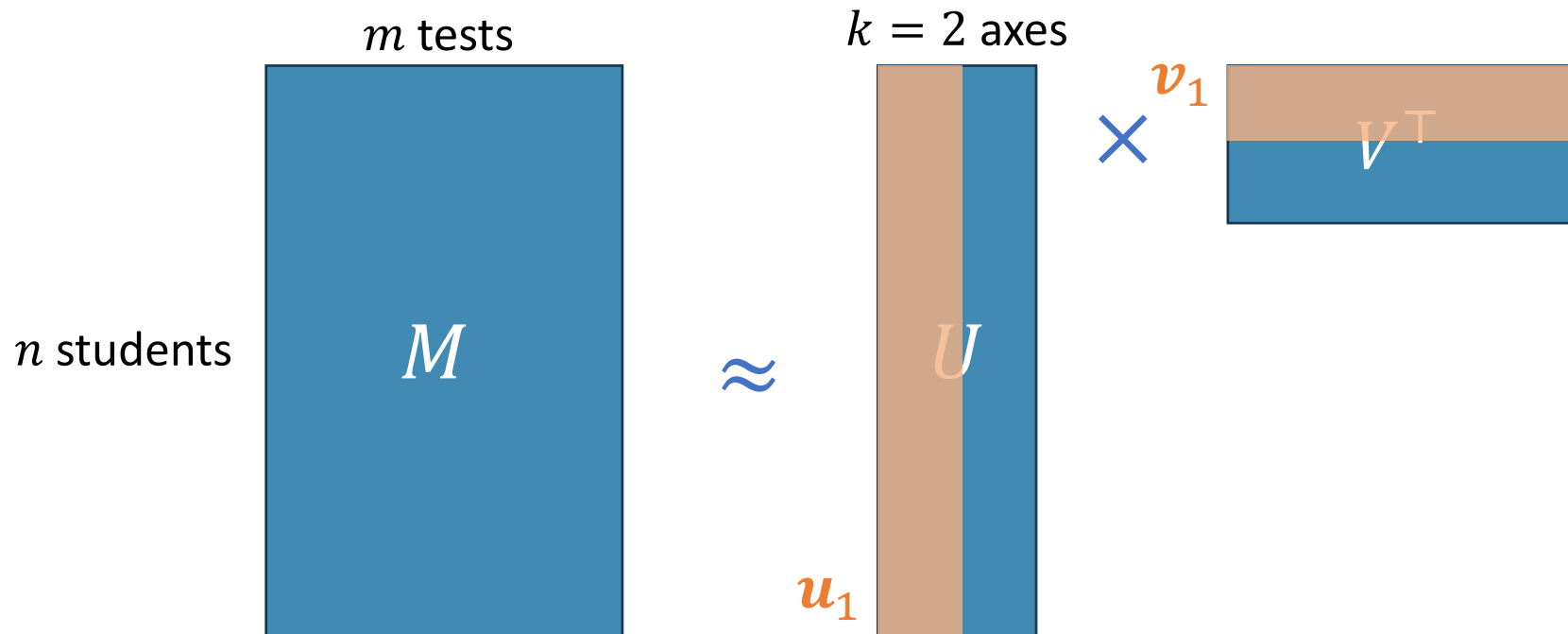


Charles Spearman
(1863-1945)

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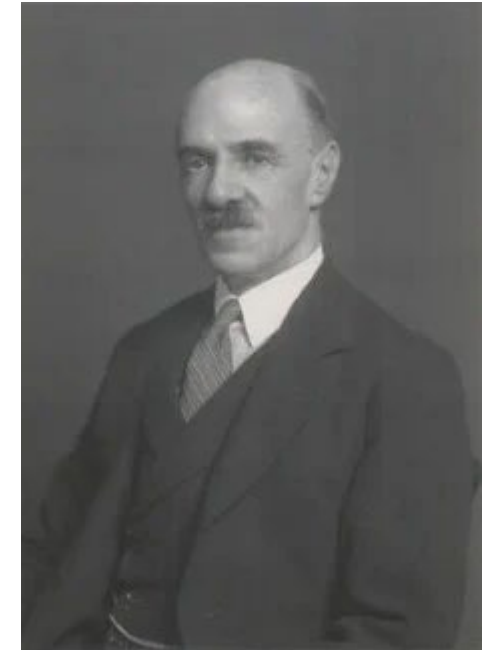
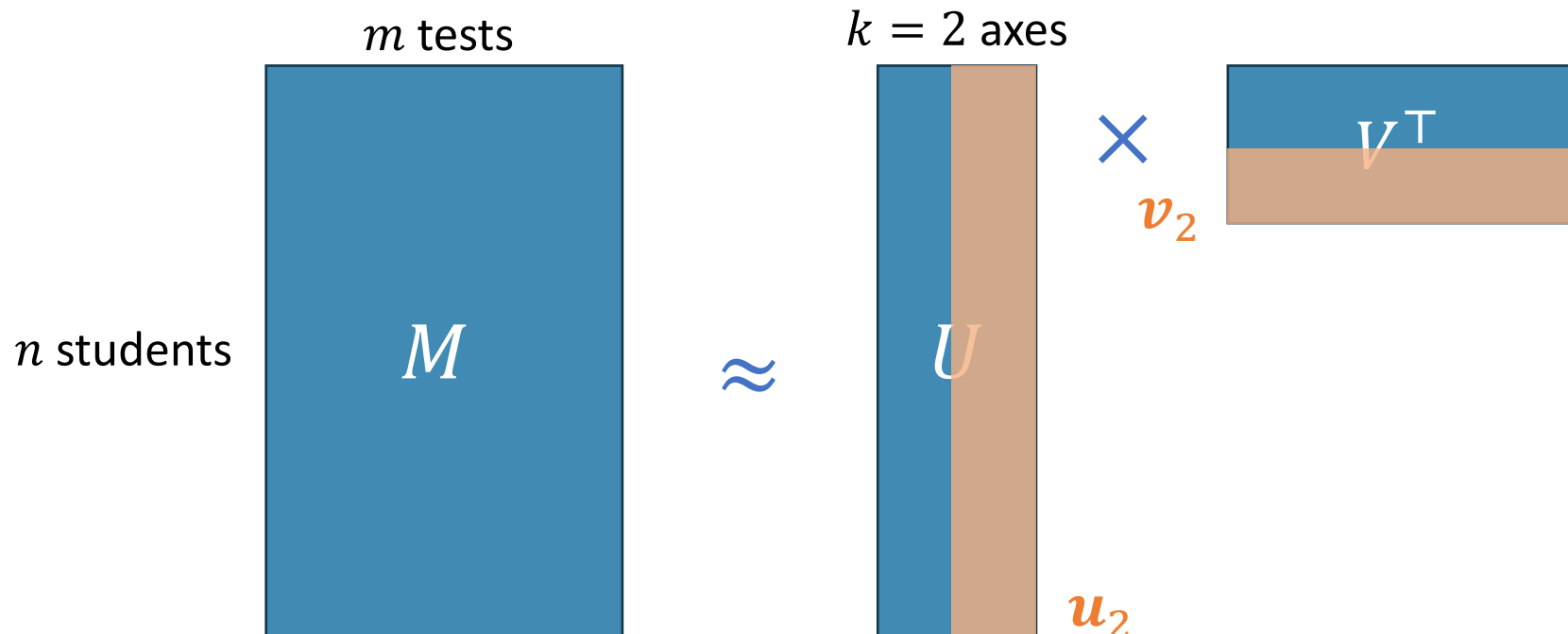


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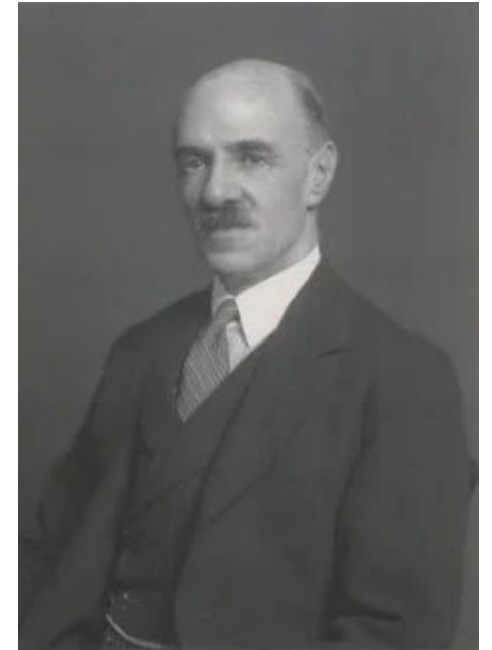
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Suppose a psychologist has the hypothesis that there are two kinds of intelligence, “verbal” and “mathematical”.

$$M = UV^{\top} = \sum_{i \in [k]} \mathbf{u}_i \mathbf{v}_i^{\top}$$

The rotation problem: this factorization is not unique

- Let $\tilde{U} \leftarrow UO$, $\tilde{V} \leftarrow VO$, where $O \in \mathbb{R}^{k \times k}$ is any orthogonal matrix
- $\tilde{U}\tilde{V}^{\top} = UOO^{\top}V^{\top} = UV^{\top}$



Charles Spearman
(1863-1945)

Detour: the Rotation Problem

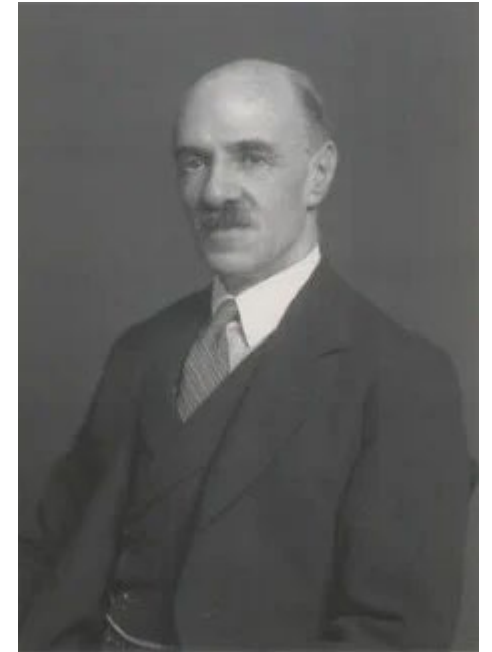
Factor analysis is a statistical method, pioneered by Charles Spearman, that explains observed correlations among many variables by modeling them as combinations of a few underlying latent factors.

Suppose a psychologist has the hypothesis that there are two kinds of intelligence, “verbal” and “mathematical”.

$$M = UV^T = \sum_{i \in [k]} \mathbf{u}_i \mathbf{v}_i^T$$

The rotation problem: this factorization is not unique

- Unless we put some additional assumptions, such as $\text{rank}(M) = 1$, $\{\mathbf{u}_k\}$ and $\{\mathbf{v}_k\}$ are orthogonal, $\{\mathbf{u}_k, \mathbf{v}_k\}$ only have non-negative entries...



Charles Spearman
(1863-1945)

The Method of Moments

Third moment ($d \times d \times d$ tensor)

Topic model

$$\Pr[x_1, x_2] = \sum_{i=1}^k w_i \boldsymbol{\mu}_i \otimes \boldsymbol{\mu}_i$$
$$\Pr[x_1, x_2, x_3] = \sum_{i=1}^k w_i \boldsymbol{\mu}_i \otimes \boldsymbol{\mu}_i \otimes \boldsymbol{\mu}_i$$

$O(d^3)$ nums, $dk + k$ vars. Possibly identifiable?

The Method of Moments

Third moment ($d \times d \times d$ tensor)

Topic model

$$\Pr[x_1, x_2] = \sum_{i=1}^k w_i \boldsymbol{\mu}_i \otimes \boldsymbol{\mu}_i$$

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Model A: $w_1 = 1/3, \boldsymbol{\mu}_1 = \begin{bmatrix} 1/6 \\ 5/6 \end{bmatrix}$ and $w_2 = 2/3, \boldsymbol{\mu}_2 = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}$

$$\Pr_A[x_1 = x_2 = x_3 = 1] = 43/216$$

Model B: $w_1 = 2/3, \boldsymbol{\mu}_1 = \begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix}$ and $w_2 = 1/3, \boldsymbol{\mu}_2 = \begin{bmatrix} 5/6 \\ 1/6 \end{bmatrix}$

$$\Pr_B[x_1 = x_2 = x_3 = 1] = 47/216$$

Detour: Tensor and Linear Algebra

- A **rank one**, **third-order** tensor T is the tensor product of three vectors $\mathbf{u} \in \mathbb{R}^{n_1}$, $\mathbf{v} \in \mathbb{R}^{n_2}$ and $\mathbf{w} \in \mathbb{R}^{n_3}$, and its entries are

$$T_{ijk} = u_i v_j w_k, \quad i \in [n_1], j \in [n_2], k \in [n_3]$$

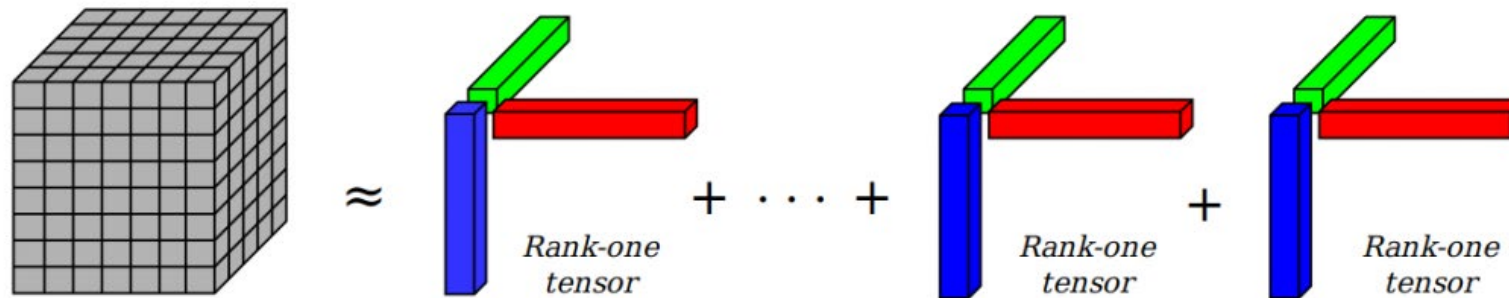
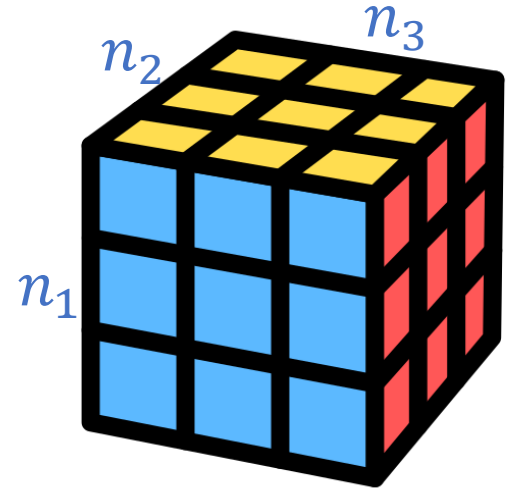
We denote it as

$$T = \mathbf{u} \otimes \mathbf{v} \otimes \mathbf{w}$$

- The **rank** of a third-order tensor T is the **smallest integer r** so that we can write

$$T = \sum_{i=1}^r \mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)} \otimes \mathbf{w}^{(i)}$$

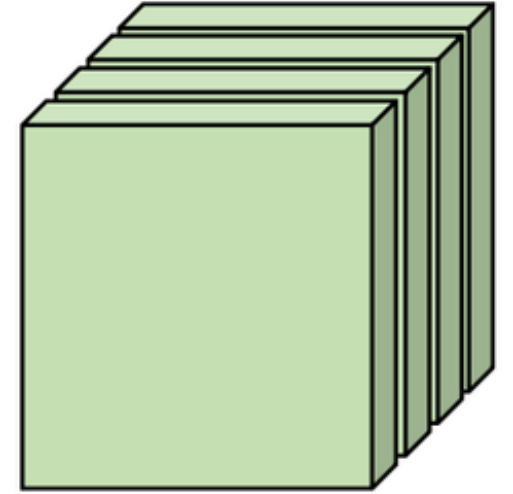
This decomposition is often called a **CP decomposition** or CANDECOMP/PARFAC decomposition



Detour: Tensor and Linear Algebra

We can view tensor T as a stacked collection of n_1 -by- n_2 matrices:

$$T(:, :, 1), T(:, :, 2), \dots, T(:, :, n_3)$$



Claim 1. If $\text{rank}(T) \leq r$, then for all $a \in [n_3]$, $\text{rank}(T(:, :, a)) \leq r$.

Proof.

$$T = \sum_{i=1}^r \mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)} \otimes \mathbf{w}^{(i)} \implies T(:, :, a) = \sum_{i=1}^r w_a^{(i)} \underbrace{(\mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)})}_{\text{rank-1 matrix}}$$

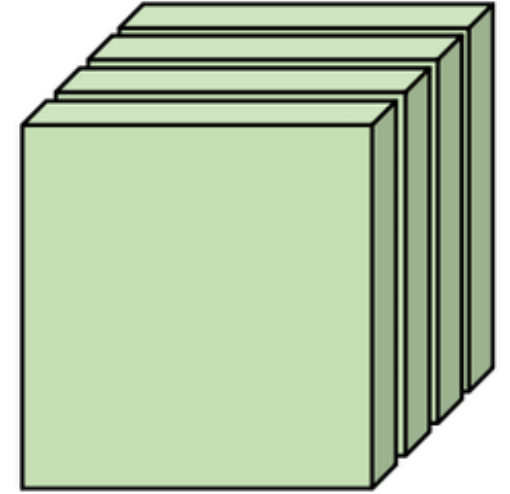
■



Detour: Tensor and Linear Algebra

We can view tensor T as a stacked collection of n_1 -by- n_2 matrices:

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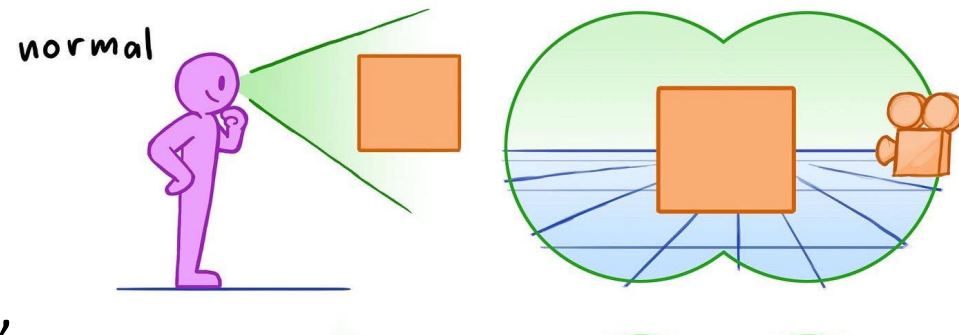
Claim 1. If $\text{rank}(T) \leq r$, then for all $a \in [n_3]$, $\text{rank}(T(:, :, a)) \leq r$.

Claim 2. Consider a tensor $T = \sum_{i=1}^r \mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)} \otimes \mathbf{w}^{(i)}$. Then, for **all** $a \in [n_3]$, we have

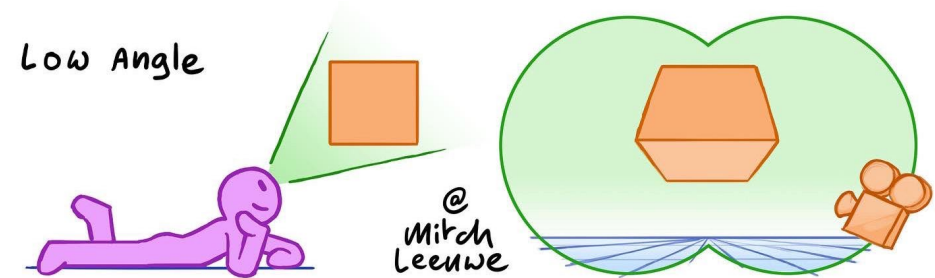
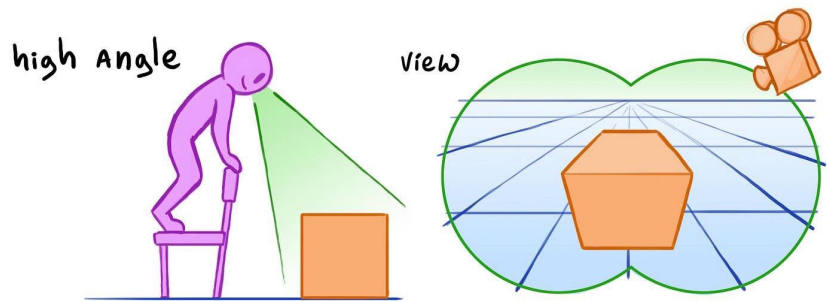
- $\text{colspan}(T(:, :, a)) \subseteq \text{span}(\{\mathbf{u}^{(i)}\})$
- $\text{rowspan}(T(:, :, a)) \subseteq \text{span}(\{\mathbf{v}^{(i)}\})$

Detour: Tensor and Linear Algebra

- **Matrix:** single “view” of $\{\mathbf{u}^{(i)}\}$ and $\{\mathbf{v}^{(i)}\}$



- **Tensor:** multiple “views”



Detour: The Trouble with Tensors

Many features of matrices that we take for granted simply **do not hold** for tensors

- For a matrix M , $\text{rank}(M) = \dim(\text{colspan}(M)) = \dim(\text{rowspan}(M))$
- For a tensor, the dimension of the span of its columns may not equal to the dimension of the span of its rows
 - Consider a tensor T of size $(n_1, n_2, n_3) = (k^2, k, k)$ such that $\{T(i, :, :)\}_{i \in [k^2]}$ are the columns of the $k^2 \times k^2$ identity matrix
 - Then, $\dim(\text{span}\{T(1, :, :), \dots, T(k^2, :, :)\}) = k^2$
 - However, $\dim(\text{span}\{T(:, 1, :), \dots, T(:, k, :)\}) \leq k$

Detour: The Trouble with Tensors

Many features of matrices that we take for granted simply **do not hold** for tensors

- The rank of a matrix will not change if we move from $\mathbb{R}$ to $\mathbb{C}$
- Tensor rank depends on the field
 - Consider the following tensor

$$T = \left[\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \right] = \frac{1}{2} \begin{bmatrix} 1 \\ -i \end{bmatrix} \otimes \begin{bmatrix} 1 \\ i \end{bmatrix} \otimes \begin{bmatrix} 1 \\ -i \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ i \end{bmatrix} \otimes \begin{bmatrix} 1 \\ -i \end{bmatrix} \otimes \begin{bmatrix} 1 \\ i \end{bmatrix}$$

- $\text{rank}_{\mathbb{C}}(T) = 2$
- $\text{rank}_{\mathbb{R}}(T) = 3$

Detour: The Trouble with Tensors

Many features of matrices that we take for granted simply **do not hold** for tensors

- If a matrix has rank r , then there is a limit of how well any rank $r' < r$ matrix can approximate it ([Eckart–Young–Mirsky theorem](#))
- There are tensors of rank 3, but which are **arbitrarily close** to tensors of rank 2
 - Consider the following tensor

$$T = \left[\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right]$$

- It can be approximated by

$$\left[\begin{bmatrix} 0 & 1 \\ 1 & 1/n \end{bmatrix}, \begin{bmatrix} 1 & 1/n \\ 1/n & 1/n^2 \end{bmatrix} \right] = \begin{bmatrix} 1 \\ 1/n \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 1/n \end{bmatrix} \otimes \begin{bmatrix} n \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} n \\ 0 \end{bmatrix}$$

- The **border rank** of T is the smallest number r such that $\forall \epsilon > 0, \exists T'$ of rank $\leq r$ such that T' is entry-wise ϵ close to T

Detour: The Trouble with Tensors

Computationally, basic linear algebraic primitives are intractable for tensors.

- **Hillar-Lim '13:** Most tensor problems are **NP-hard**

Eigenvalue over $\mathbb{R}$	NP-hard (Theorem 1.3)
Approximating Eigenvector over $\mathbb{R}$	NP-hard (Theorem 1.5)
Symmetric Eigenvalue over $\mathbb{R}$	NP-hard (Theorem 9.3)
Approximating Symmetric Eigenvalue over $\mathbb{R}$	NP-hard (Theorem 9.6)
Singular Value over $\mathbb{R}, \mathbb{C}$	NP-hard (Theorem 1.7)
Symmetric Singular Value over $\mathbb{R}$	NP-hard (Theorem 10.2)
Approximating Singular Vector over $\mathbb{R}, \mathbb{C}$	NP-hard (Theorem 6.3)
Spectral Norm over $\mathbb{R}$	NP-hard (Theorem 1.10)
Symmetric Spectral Norm over $\mathbb{R}$	NP-hard (Theorem 10.2)
Approximating Spectral Norm over $\mathbb{R}$	NP-hard (Theorem 1.11)
Nonnegative Definiteness	NP-hard (Theorem 11.2)
Best Rank-1 Approximation	NP-hard (Theorem 1.13)
Best Symmetric Rank-1 Approximation	NP-hard (Theorem 10.2)
Rank over $\mathbb{R}$ or $\mathbb{C}$	NP-hard (Theorem 8.2)

Tensor Decomposition

Given a tensor $T \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ such that

$$T = \sum_{i=1}^r \mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)} \otimes \mathbf{w}^{(i)}$$

Our goal is to recover the set of factors $\{(\mathbf{u}^{(i)}, \mathbf{v}^{(i)}, \mathbf{w}^{(i)})\}_{i \in [r]}$

- There are some **symmetries** in this decomposition
- $\{(\mathbf{u}^{(i)}, \mathbf{v}^{(i)}, \mathbf{w}^{(i)})\}$ and $\{(\hat{\mathbf{u}}^{(i)}, \hat{\mathbf{v}}^{(i)}, \hat{\mathbf{w}}^{(i)})\}$ are **equivalent** if there exists a permutation $\pi : [r] \rightarrow [r]$ such that

$$\mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)} \otimes \mathbf{w}^{(i)} = \hat{\mathbf{u}}^{(i)} \otimes \hat{\mathbf{v}}^{(i)} \otimes \hat{\mathbf{w}}^{(i)}, \quad \forall i \in [r]$$

When are the factors of T determined up to equivalence?

Tensor Decomposition

Theorem (Harshman, Jennrich '70). Suppose the following conditions hold:

- 1) $\{u_i\}$ are linearly independent
- 2) $\{v_i\}$ are linearly independent
- 3) $n_3 \geq 2$ and no two w_i, w_j are collinear

Then the factors are uniquely determined up to equivalence, and there is a **polynomial time algorithm** to find them

Tensor decomposition: Jennrich's algorithm

1. Choose $\mathbf{a}, \mathbf{b} \in \mathbb{R}^{n_3}$ uniformly random unit vectors
2. Set

$$T(\mathbf{a}) := \sum_{i \in [n_3]} a_i T(:, :, i) \quad \text{and} \quad T(\mathbf{b}) := \sum_{i \in [n_3]} b_i T(:, :, i) \quad \text{tensor contraction}$$

3. Compute $A := T(\mathbf{a})T(\mathbf{b})^{-1}$ and $B := (T(\mathbf{a})^{-1}T(\mathbf{b}))^\top$
4. Let $\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(r)}$ be eigenvectors of A with eigenvalues $\lambda_1, \dots, \lambda_r$
5. Let $\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(r)}$ be eigenvectors of B with eigenvalues $\lambda_1^{-1}, \dots, \lambda_r^{-1}$
6. Solve linear system to recover $\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(r)}$:

$$T = \sum_{i=1}^r \mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)} \otimes \mathbf{w}^{(i)}$$

Tensor decomposition: Jennrich's algorithm

- Let $U \in \mathbb{R}^{n_1 \times r}$ and $V \in \mathbb{R}^{n_2 \times r}$ be the matrices with columns $\{\mathbf{u}^{(i)}\}$ and $\{\mathbf{v}^{(i)}\}$
- Let $D_a, D_b \in \mathbb{R}^{r \times r}$ be diagonal matrices with entries $\langle \mathbf{a}, \mathbf{w}^{(i)} \rangle$ and $\langle \mathbf{b}, \mathbf{w}^{(i)} \rangle$

Lemma. We have that

$$T(\mathbf{a}) = UD_aV^\top, \quad T(\mathbf{b}) = UD_bV^\top$$

Proof.

$$\begin{aligned} T(\mathbf{a}) &= \sum_{i \in [n_3]} a_i T(:, :, i) = \sum_{i \in [n_3]} a_i \sum_{j \in [r]} (\mathbf{u}^{(j)} \otimes \mathbf{v}^{(j)}) (\mathbf{w}^{(j)})_i \\ &= \sum_{j \in [r]} \mathbf{u}^{(j)} \otimes \mathbf{v}^{(j)} \langle \mathbf{a}, \mathbf{w}^{(j)} \rangle \\ &= UD_aV^\top \end{aligned}$$



Tensor decomposition: Jennrich's algorithm

- Let $U \in \mathbb{R}^{n_1 \times r}$ and $V \in \mathbb{R}^{n_2 \times r}$ be the matrices with columns $\{\mathbf{u}^{(i)}\}$ and $\{\mathbf{v}^{(i)}\}$
- Let $D_{\mathbf{a}}, D_{\mathbf{b}} \in \mathbb{R}^{r \times r}$ be diagonal matrices with entries $\langle \mathbf{a}, \mathbf{w}^{(i)} \rangle$ and $\langle \mathbf{b}, \mathbf{w}^{(i)} \rangle$

Lemma. We have that

$$T(\mathbf{a}) = UD_{\mathbf{a}}V^{\top}, \quad T(\mathbf{b}) = UD_{\mathbf{b}}V^{\top}$$

- Consider the following **generalized eigenvalue problem**:

$$T(\mathbf{a})\mathbf{v} = \lambda_v T(\mathbf{b})\mathbf{v}, \quad UD_{\mathbf{a}}V^{\top}\mathbf{v} = \lambda_v UD_{\mathbf{b}}V^{\top}\mathbf{v}$$

- $\{\mathbf{v}^{(i)}\}$ are eigenvectors, **suppose** they are orthonormal. Then,

$$\begin{aligned} UD_{\mathbf{a}}V^{\top}\mathbf{v}^{(i)} &= UD_{\mathbf{a}}\mathbf{e}_i = \langle \mathbf{a}, \mathbf{w}^{(i)} \rangle U\mathbf{e}_i = \langle \mathbf{a}, \mathbf{w}^{(i)} \rangle \mathbf{u}^{(i)} \\ UD_{\mathbf{b}}V^{\top}\mathbf{v}^{(i)} &= UD_{\mathbf{b}}\mathbf{e}_i = \langle \mathbf{b}, \mathbf{w}^{(i)} \rangle U\mathbf{e}_i = \langle \mathbf{b}, \mathbf{w}^{(i)} \rangle \mathbf{u}^{(i)} \end{aligned}$$

$$T(\mathbf{a})\mathbf{v}^{(i)} = \lambda_i T(\mathbf{b})\mathbf{v}^{(i)}, \quad \lambda_i = \frac{\langle \mathbf{b}, \mathbf{w}^{(i)} \rangle}{\langle \mathbf{a}, \mathbf{w}^{(i)} \rangle}$$

Tensor decomposition: Jennrich's algorithm

- Let $U \in \mathbb{R}^{n_1 \times r}$ and $V \in \mathbb{R}^{n_2 \times r}$ be the matrices with columns $\{\mathbf{u}^{(i)}\}$ and $\{\mathbf{v}^{(i)}\}$
- Let $D_{\mathbf{a}}, D_{\mathbf{b}} \in \mathbb{R}^{r \times r}$ be diagonal matrices with entries $\langle \mathbf{a}, \mathbf{w}^{(i)} \rangle$ and $\langle \mathbf{b}, \mathbf{w}^{(i)} \rangle$

$$T(\mathbf{a})\mathbf{v}^{(i)} = \lambda_i T(\mathbf{b})\mathbf{v}^{(i)}, \quad \lambda_i = \frac{\langle \mathbf{b}, \mathbf{w}^{(i)} \rangle}{\langle \mathbf{a}, \mathbf{w}^{(i)} \rangle}$$

- If all λ_i are distinct, then solving this generalized eigenvalue problem can uniquely recover the generalized eigenvectors $\{\mathbf{v}^{(i)}\}$ (up to a rescaling)

Tensor decomposition: Jennrich's algorithm

Lemma. Almost surely, the columns of U and V are the unique eigenvectors corresponding to non-zero eigenvalues of $T(\mathbf{a})T(\mathbf{b})^{-1}$ and $(T(\mathbf{a})^{-1}T(\mathbf{b}))^\top$ respectively. Moreover, the eigenvalue corresponding to $\mathbf{u}^{(i)}$ is the reciprocal of the eigenvalue corresponding to $\mathbf{v}^{(i)}$

Proof.

- Using the formula for $T(\mathbf{a})$ and $T(\mathbf{b})$, we have

$$T(\mathbf{a})T(\mathbf{b})^{-1} = UD_aV^\top(UD_bV^\top)^{-1} = UD_aV^\top(V^\top)^{-1}D_b^{-1}U^{-1} = U\mathbf{D}_a\mathbf{D}_b^{-1}U^{-1}$$
$$(T(\mathbf{a})^{-1}T(\mathbf{b}))^\top = ((V^\top)^{-1}D_a^{-1}U^{-1}UD_bV^\top)^\top = ((V^\top)^{-1}D_a^{-1}D_bV^\top)^\top = V\mathbf{D}_b\mathbf{D}_a^{-1}V^{-1}$$

- The diagonal entries of $D_aD_b^{-1}$ is $\langle \mathbf{a}, \mathbf{w}^{(i)} \rangle / \langle \mathbf{b}, \mathbf{w}^{(i)} \rangle$
- Since every pair of vectors in $\{\mathbf{w}^{(i)}\}$ are not collinear and $\mathbf{a}, \mathbf{b}$ are **random** vectors, almost surely all diagonal entries are non-zero and distinct
- Thus, $\{\mathbf{u}^{(i)}\}$ and $\{\mathbf{v}^{(i)}\}$ are unique eigenvectors, and they are matched by reciprocal eigenvalues



Tensor decomposition: Jennrich's algorithm

- To recover $w^{(i)}$, we solve a linear system:

$$T = \sum_{i=1}^r \mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)} \otimes \mathbf{w}^{(i)}$$

- The ground-truth $\{\mathbf{w}^{(i)}\}$ is a solution
- We need to show that this is the only solution
- Observe that this big linear system can be decomposed as n_3 independent smaller linear systems:

$$T(:, :, a) = \sum_{i=1}^r \mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)} w_a^{(i)}, \quad a \in [n_3]$$

$$[\mathbf{u}^{(1)} \otimes \mathbf{v}^{(1)} \quad \dots \quad \mathbf{u}^{(r)} \otimes \mathbf{v}^{(r)}] \begin{bmatrix} w_a^{(1)} \\ \vdots \\ w_a^{(r)} \end{bmatrix} = T(:, :, a)$$

Tensor decomposition: Jennrich's algorithm

Lemma. $\{\mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)}\}_{i \in [r]}$ are linearly independent

Proof.

- Suppose there exists not all zero $\alpha_1, \dots, \alpha_r$ such that

$$\sum_{i=1}^r \alpha_i \mathbf{u}^{(i)} \otimes \mathbf{v}^{(i)} = \mathbf{0}$$

- Wlog, assume $\alpha_1 \neq 0$. Since $\{\mathbf{v}^{(i)}\}$ are linearly independent, we can find $\mathbf{a} \in \mathbb{R}^{n_2}$ such that

$$\langle \mathbf{a}, \mathbf{v}^{(1)} \rangle \neq 0, \quad \langle \mathbf{a}, \mathbf{v}^{(i)} \rangle = 0 \quad \forall i \neq 1$$

$$\mathbf{0} = \sum_{i=1}^r \alpha_i \mathbf{u}^{(i)} \otimes \langle \mathbf{a}, \mathbf{v}^{(i)} \rangle = \alpha_1 \mathbf{u}^{(1)} \langle \mathbf{a}, \mathbf{v}^{(1)} \rangle$$

- This is a contradiction since $\text{RHS} \neq \mathbf{0}$

